

# MATHEMATICS FOR DATA SCIENCE & MACHINE LEARNING

(Vol. 3) Comprehensive Practical Question Bank: Questions 401–600

This workbook acts as Volume 3 of your advanced multi-disciplinary math suite, containing an entirely original set of 200 rigorous practice problems across Linear Algebra, Probability, Statistics, and Differential Calculus. Engineered with concrete, realistic metrics, matrices, and datasets, these questions span foundational, intermediate, and advanced applied tracks explicitly customized to fortify your algorithmic intuition for real-world AI and deep learning pipelines.

## 1. LINEAR ALGEBRA (QUESTIONS 401–450)

### A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 401–415)

**401.** Given the structural column vectors  $\mathbf{a} = [3, -2, 6]^T$  and  $\mathbf{b} = [1, 4, -2]^T$ , compute the precise linear combination  $3\mathbf{a} - 2\mathbf{b}$  and evaluate its Euclidean  $L_2$  norm.

**402.** Calculate the dot product of two attention-head projection keys  $\mathbf{u} = [4, -1, 0, 2]$  and  $\mathbf{v} = [-2, 5, 3, 1]$ . State explicitly whether the vectors form an acute, obtuse, or right angle.

**403.** Find the Manhattan ( $L_1$ ) and Chebyshev ( $L_\infty$ ) norms of the gradient optimization tracking discrepancy vector  $\Delta\mathbf{w} = [-0.12, 0.45, -0.08, 0.95, -0.34]$ .

**404.** Compute the exact cosine similarity score between two user-preference latent representation vectors:  $\mathbf{x} = [1, 5, 2]$  and  $\mathbf{y} = [3, 0, 4]$ .

**405.** Find the resulting matrix-vector product  $\mathbf{y} = \mathbf{W}\mathbf{x}$  given the weight matrix  $\mathbf{W}$  and input vector  $\mathbf{x}$ :

$$\mathbf{W} = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \\ 5 & -2 & -3 \end{bmatrix}$$

$$\mathbf{x} = [2, 3, -1]^T$$

**406.** For the following two dense neural network transformation matrices, compute the structural matrix product  $\mathbf{C} = \mathbf{AB}$ :

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 0 & 4 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 4 & 0 \end{bmatrix}$$

**407.** Verify if matrix multiplication is commutative for the following square token matrices by explicitly evaluating both  $AB$  and  $BA$ :

$$\begin{aligned} A &= \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \\ B &= \begin{bmatrix} 5 & 0 \\ 2 & -1 \end{bmatrix} \end{aligned}$$

**408.** Demonstrate that the transpose distribution property  $(CD)^T = D^T C^T$  holds exactly using the given mini-matrices:

$$\begin{aligned} C &= \begin{bmatrix} 1 & 4 \\ -2 & 3 \end{bmatrix} \\ D &= \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix} \end{aligned}$$

**409.** Find the determinant of the  $2 \times 2$  spatial projection transformation matrix  $M = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ . Is  $M$  singular or non-singular?

**410.** Compute the analytical matrix inverse  $W^{-1}$  for the  $2 \times 2$  layer weight operator:

$$W = \begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}$$

**411.** Find the matrix trace of the square feature variance-covariance matrix  $\Sigma$  given by:

$$\Sigma = \begin{bmatrix} 6.2 & 0.8 & 0.0 \\ 0.8 & 4.7 & -1.5 \\ 0.0 & -1.5 & 7.3 \end{bmatrix}$$

**412.** Determine whether the following three extracted text feature vectors are linearly independent or dependent within their embedding space:  $x_1 = [1, 2, 0]$ ,  $x_2 = [0, 1, 3]$ , and  $x_3 = [2, 5, 3]$ .

**413.** Determine the structural rank of matrix  $A$  by mapping it directly into its row echelon form:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

**414.** Project the support vector machine boundary data point  $x = [5, 2]^T$  onto the linear subspace spanned by the unit vector  $v = [0.6, 0.8]^T$ . Find the exact scalar projection value.

**415.** Determine whether the spatial rotation matrix  $R$  acts as a true orthogonal matrix by verifying if  $R^T R = I$ :

$$R = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$$

## B. INTERMEDIATE LEVEL (QUESTIONS 416–435)

**416.** Solve the following linear system  $Ax = b$  via Gaussian Elimination, representing a 3-factor production supply pipeline optimization framework:

$$\begin{aligned} x + y + 2z &= 8 \\ -x + 2y + z &= 5 \\ 2x + y - z &= 1 \end{aligned}$$

**417.** Find the determinant of the  $3 \times 3$  layer scaling matrix using cofactor expansion down the second column:

$$A = \begin{bmatrix} 3 & 0 & 2 \\ 1 & 5 & -2 \\ 4 & 1 & 3 \end{bmatrix}$$

**418.** Find the mathematical inverse of the  $3 \times 3$  block matrix  $B$  utilizing the Gauss-Jordan elimination row protocol:

$$B = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 2 \\ 3 & 0 & 1 \end{bmatrix}$$

**419.** Compute the exact eigenvalues and corresponding analytical eigenvectors for the  $2 \times 2$  feature transformation matrix  $M = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ .

**420.** Given the symmetric covariance structure  $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ , determine its complete orthogonal diagonalization components  $P$  and  $D$  such that  $A = PDP^T$ .

**421.** Compute the total Frobenius norm of the given  $3 \times 2$  parameter neural weight update discrepancy matrix  $\Delta W$ :

$$\Delta W = \begin{bmatrix} 0.4 & -1.2 \\ 0.8 & 0.3 \\ -0.5 & 0.6 \end{bmatrix}$$

**422.** Given a small matrix  $X$  of 3 sequential observations and 2 features, compute the uncentered Gram matrix  $G = XX^T$ :

$$X = \begin{bmatrix} 2 & 1 \\ 1 & 4 \\ 3 & 0 \end{bmatrix}$$

**423.** Test whether the specific quadratic form loss function  $q(x) = 3x_1^2 - 2x_1x_2 + 4x_2^2$  is strictly positive definite using Sylvester's criterion on principal minors.

**424.** A custom vector has coordinates  $x_B = [2, 4]^T$  relative to the alternative basis  $B = \{[3, 1]^T, [-2, 5]^T\}$ . Determine its precise coordinate representation in the standard basis.

**425.** Compute the Hadamard element-wise product  $U \odot V$  and the standard matrix product  $UV$  for the given network dropout masks:

$$\begin{aligned} U &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \\ V &= \begin{bmatrix} 0.8 & 3 \\ 0.0 & 2.5 \end{bmatrix} \end{aligned}$$

**426.** Apply the classical Gram-Schmidt orthonormalization process to transform the linearly independent basis vectors  $u_1 = [0, 1, 1]$  and  $u_2 = [2, 1, 0]$  into an orthonormal basis.

**427.** Determine the structural basis and true dimension of the null space (kernel) for the system matrix:

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 6 & -2 \\ -3 & -9 & 3 \end{bmatrix}$$

**428.** Compute the system condition number  $\kappa(A)$  in the  $L_\infty$  norm for the highly sensitive linear scaling matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2.004 \end{bmatrix}$$

**429.** Find the explicit LU Decomposition ( $A = LU$ ) without pivoting for the layer matrix:

$$A = \begin{bmatrix} 3 & 2 \\ 12 & 11 \end{bmatrix}$$

**430.** A directional pipeline network maps its links via the adjacency matrix  $A = [[0, 0, 1], [1, 0, 0], [1, 1, 0]]$ . Compute  $A^2$  to establish the number of distinct 2-step pathways connecting the network nodes.

**431.** Find the spanning basis for the row space and the column space of the following structural feature matrix:

$$M = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 0 & 4 \\ 3 & 3 & 9 \end{bmatrix}$$

**432.** Compute the exact Moore-Penrose pseudo-inverse  $A^+$  for the single-row vector matrix  $A = [2, 5]$ .

**433.** Express the standard space vector  $v = [5, 11]$  as an explicit linear combination of the alternative coordinate systems defined by basis vectors  $e_1 = [2, 3]$  and  $e_2 = [1, 4]$ .

**434.** An affine transformation maps data by scaling a factor of 3 along the x-axis and translating the result by vector  $b = [-2, 5]^T$ . Formulate this mapping as  $Ax + b$  and evaluate it directly for  $x = [4, 1]^T$ .

**435.** Show that if a projection operator matrix satisfies  $P = P^T$  and  $P = P^2$ , then the modified vector  $(I - P)x$  is perfectly orthogonal to  $Px$  for any arbitrary input vector  $x$ .

### C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 436–450)

**436.** Compute the complete Singular Value Decomposition (SVD),  $A = U\Sigma V^T$ , for the non-square rectangular data matrix:

$$A = \begin{bmatrix} 2 & 2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

**437.** A zero-mean centered training mini-dataset matrix  $X$  contains 4 samples mapped across 2 distinct features:

$$X = \begin{bmatrix} [-1, & 2], \\ [-2, & -1], \\ [2, & 0], \\ [1, & -1] \end{bmatrix}$$

Compute the empirical covariance matrix  $\Sigma = (1/4)X^T X$  and determine its principal component loading directions via its eigenvectors.

**438.** Find the optimal least-squares parameter solution vector  $w = (X^T X)^{-1} X^T y$  for the linear regression design matrix  $X$  and target labels  $y$ :

$$X = \begin{bmatrix} [1, & 1], \\ [1, & 2], \\ [1, & 3] \end{bmatrix} \\ y = [2, 5, 5]^T$$

**439.** Project the out-of-sample data point  $x = [3, 2]$  onto the primary top principal component loading direction of the feature covariance matrix  $\Sigma = [[4, 2], [2, 1]]$ .

**440.** Compute the Rank-1 optimal low-rank matrix approximation  $A_1$  via truncated SVD rules for the following diagonal scaling layout:

$$A = \begin{bmatrix} [5, & 0], \\ [0, & 2] \end{bmatrix}$$

**441.** Find the exact Mahalanobis distance between sample target point  $x = [4, 3]^T$  and cluster centroid  $\mu = [2, 1]^T$  given the inverse data covariance structure:

$$\Sigma^{-1} = \begin{bmatrix} [1.5, & -0.5], \\ [-0.5, & 1.0] \end{bmatrix}$$

**442.** Given a graph adjacency matrix  $A = [[0, 1, 0], [1, 0, 1], [0, 1, 0]]$  and its corresponding diagonal degree matrix  $D$ , compute the structural unnormalized graph Laplacian matrix  $L = D - A$ .

**443.** Derive the analytical gradient vector with respect to parameter vector  $w$  for the weighted ridge constraint objective  $f(w) = w^T D w$ , where  $D$  is a constant diagonal weight matrix.

**444.** Compute the explicit Moore-Penrose pseudo-inverse matrix  $X^+$  for the full-column-rank design matrix:

$$X = \begin{bmatrix} [1, & 0], \\ [1, & 1], \\ [1, & 2] \end{bmatrix}$$

**445.** Prove rigorously that the eigenvalues of a symmetric positive definite matrix must all be strictly positive real numbers.

**446.** A steady-state Markov page routing probability satisfies the vector invariant  $vP = v$ . Find the normalized probability vector  $v$  given the stochastic routing matrix:

$$P = \begin{bmatrix} [0.6, & 0.4], \\ [0.3, & 0.7] \end{bmatrix}$$

- 447.** Using a Radial Basis Function (RBF) kernel matrix calculation  $K(x,y) = \exp(-\gamma\|x - y\|^2)$  with scaling factor  $\gamma = 0.25$ , calculate the kernel similarity score between local points  $x = [2, 4]$  and  $y = [1, 2]$ .
- 448.** Determine the structural data whitening transformation matrix  $W = \Sigma^{-1/2}$  for an image dataset whose current feature covariance is given by  $\Sigma = \begin{bmatrix} 9 & 0 \\ 0 & 16 \end{bmatrix}$ .
- 449.** Compute the Kronecker product  $A \otimes B$  for the mini-batch spatial operators:

$$\begin{aligned} A &= \begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix} \\ B &= \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix} \end{aligned}$$

- 450.** In an optimization context, show that the spectral radius (maximum absolute eigenvalue) of the iteration matrix  $B$  must be strictly less than 1 for the linear numerical system  $x_{k+1} = Bx_k + c$  to converge cleanly to its fixed-point solution.

## 2. PROBABILITY (QUESTIONS 451–500)

### A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 451–465)

- 451.** An edge API router channels 80% of server calls to Cluster Alpha and 20% to Cluster Beta. Cluster Alpha throws an exception on 2% of calls, while Cluster Beta drops 5% of calls. Find the total probability that an incoming request errors out.
- 452.** Given probabilities  $P(A) = 0.5$ ,  $P(B) = 0.4$ , and  $P(A \cap B) = 0.20$ , compute conditional score  $P(A|B)$  and state whether events  $A$  and  $B$  are independent.
- 453.** In an enterprise fraud monitoring framework, 3% of incoming transactions are illicit. An automated anomaly detector flags 95% of fraudulent transactions and 4% of legitimate ones. Calculate the posterior probability that a transaction flagged as an anomaly is actually fraudulent.
- 454.** A discrete random feature score  $X$  displays the following Probability Mass Function (PMF):

|             |     |     |     |     |
|-------------|-----|-----|-----|-----|
| <b>X</b>    | 0   | 2   | 4   | 8   |
| <b>P(X)</b> | 0.3 | 0.4 | 0.2 | 0.1 |

Find the exact expected value  $E[X]$ .

- 455.** For the discrete random feature variable  $X$  detailed in Question 454, compute the exact variance  $\text{Var}(X)$  and standard deviation  $\sigma$ .
- 456.** A lead conversion rate tracking model indicates  $p = 0.4$ . Using the Binomial distribution formula, calculate the probability of securing exactly 4 conversions out of 7 independent user clicks.
- 457.** The number of system warnings logged by a telemetry script follows a Poisson distribution averaging 2.0 warnings per hour. Compute the probability of seeing exactly 4 warnings in a given hour.
- 458.** A data pipeline cleaning delay  $X$  is uniformly distributed between 10 and 40 seconds. Find the probability that a randomly sampled validation task requires between 15 and 30 seconds.

- 459.** Database read latencies are normally distributed with a mean  $\mu = 150$  ms and standard deviation  $\sigma = 20$  ms. Calculate the standard Z-score corresponding to a sensor reading of 185 ms.
- 460.** Using standard normal lookup statistics where  $\Phi(1.75) = 0.9599$ , find the probability that a randomly sampled database query runs at a latency below 185 ms based on the parameters in Question 459.
- 461.** A web crawling script has a probability  $p = 0.08$  of extracting a broken webpage. What is the expected number of pages parsed before hitting the first broken page?
- 462.** If three independent training pipeline stages have individual success probabilities of 0.9, 0.85, and 0.95, what is the combined probability that all three stages complete successfully?
- 463.** Express the statistical Recall metric of a classification validation test as a conditional probability involving true positive states and conditional label space states.
- 464.** A Wi-Fi router drops packets with a 1% probability. Find the probability that the first dropped packet occurs on exactly the 5th packet transmission.
- 465.** Prove the Law of Total Probability analytically for a continuous sample space partitioned into three mutually exclusive scenarios  $B_1$ ,  $B_2$ , and  $B_3$ .

## B. INTERMEDIATE LEVEL (QUESTIONS 466–485)

**466.** A medical imaging computer vision model displays 98% sensitivity and 95% specificity. The baseline condition incidence rate in the target cohort is 2%. If an individual tests positive, evaluate the exact posterior probability they have the underlying condition.

**467.** Two random variable discrete engineering metrics  $X$  and  $Y$  reveal the following joint PMF grid:

|         | $Y = 1$ | $Y = 2$ |
|---------|---------|---------|
| $X = 1$ | 0.20    | 0.30    |
| $X = 2$ | 0.40    | 0.10    |

Compute the individual marginal PMF values  $P(X)$  and  $P(Y)$  for all states.

- 468.** Using the joint probability table in Question 467, compute the conditional expectation value  $E[X | Y = 1]$ .
- 469.** Calculate the exact statistical covariance  $Cov(X, Y)$  using the joint distribution provided in Question 467.
- 470.** Determine the Pearson correlation coefficient  $\rho_{X,Y}$  for the features described in the joint grid of Question 467.
- 471.** The time required to process an API query order follows an exponential layout with a mean duration of 5 minutes ( $\lambda = 0.20$ ). Calculate the probability that an incoming order takes more than 8 minutes to process.
- 472.** Given two uncorrelated random feature adjustments  $A$  and  $B$  with known variances  $Var(A) = 9$  and  $Var(B) = 16$ , determine the exact variance of the linear contrast score  $Z = 3A - 2B$ .
- 473.** A continuous probability density curve operates via the function  $f(x) = k(2 - x)$  for  $0 \leq x \leq 2$ , and 0 elsewhere. Determine the proper normalization constant  $k$ .
- 474.** For the density curve established in Question 473, find the mathematical expected value  $E[X]$ .

- 475.** Derive the explicit Cumulative Distribution Function expression  $F(x) = P(X \leq x)$  for the bounded continuous density in Question 473.
- 476.** User connection dropouts follow a Poisson pattern at an average rate of 0.3 dropouts per minute. Find the probability of noting zero dropouts during a 15-minute network monitoring test.
- 477.** A distributed storage node relies on three independent backup hard loops, each displaying a standalone survival probability of 0.90. The system remains operational if at least one loop survives. Find the overall reliability.
- 478.** Using the Central Limit Theorem, approximate the probability that the average score of 50 independent uniform samples drawn from  $U(0, 6)$  is strictly greater than 3.4. (For  $U(0, 6)$ ,  $\mu = 3$ ,  $\sigma^2 = 36/12 = 3$ ).
- 479.** Prove analytically that the covariance between two random variables satisfies the fundamental identities relation  $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$ .
- 480.** Let independent neural latent layers be modeled normally as  $A \sim N(8, 3)$  and  $B \sim N(15, 5)$ . Define the parameters for the distribution of their difference score  $W = B - A$ .
- 481.** A production classifier model displays a 15% error rate on general test sets. If you sample 5 test items completely at random, what is the probability that at most 1 item is misclassified?
- 482.** Assume conditional naive profiles indicate  $P(X | \text{Class A}) = 0.6$  and  $P(X | \text{Class B}) = 0.3$ . If the prior probability of Class A is 0.4 and Class B is 0.6, calculate the true posterior probability  $P(\text{Class A} | X)$ .
- 483.** An extraction asset box holds 9 clean log files and 3 corrupted ones. If you draw 3 files sequentially without replacement, find the probability that exactly 1 corrupted log file is chosen.
- 484.** Compute the Shannon entropy in bits for a 4-state multinomial categorical feature distribution with state probabilities given by  $p = [0.25, 0.25, 0.25, 0.25]$ .
- 485.** Apply Chebyshev's Inequality to find the maximum possible upper-bound probability that a random feature observation deviates from its mean by more than 2.5 standard deviations.

### C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 486–500)

- 486.** A bivariate normal feature pair  $[X, Y]^T$  has a mean vector  $\mu = [2, 3]^T$  and covariance structure  $\Sigma = [[3, 1], [1, 3]]$ . Write out the complete analytical algebraic joint density function expression  $f(x, y)$ .
- 487.** For the bivariate distribution established in Question 486, derive the explicit conditional normal distribution parameters (mean and variance) for  $Y$  given the observation that  $X = 3$ .
- 488.** Compute the exact Kullback-Leibler (KL) Divergence  $D_{KL}(P \parallel Q)$  between the target categorical distribution  $P = [0.6, 0.4]$  and model approximation  $Q = [0.5, 0.5]$ .
- 489.** Prove that the KL divergence between two multivariate normal distributions  $N_0(\mu_0, \Sigma_0)$  and  $N_1(\mu_1, \Sigma_1)$  reduces to a trace and quadratic form expression when both covariance matrices are identical to the identity matrix  $\Sigma_0 = \Sigma_1 = I$ .
- 490.** In a text document multi-label framework, write out the complete joint probability formulation for a text item containing three conditionally independent tokens given the parent topic class label.



- 491.** Derive the explicit Moment Generating Function (MGF),  $M_X(t) = E[e^{tX}]$ , for a continuous random variable following an Exponential distribution layout with rate parameter  $\lambda$ .
- 492.** Using the Inverse CDF transformation rule, derive a sampling expression to generate random draws from a Pareto distribution with minimum scale parameter  $x_m$  and shape parameter  $\alpha$ , given standard uniform random variables  $U \sim \text{Uniform}(0, 1)$ .
- 493.** A Gaussian mixture profile tracks three distinct cluster density nodes with weights  $\pi = [0.1, 0.6, 0.3]$ . Given their individual component densities are  $p_1(x)$ ,  $p_2(x)$ , and  $p_3(x)$ , write out the complete marginal density equation for an arbitrary observation  $x$ .
- 494.** Derive the explicit variance matrix expression for a linear mapping vector transformation  $Y = AX$ , where  $X$  is a random vector with covariance matrix  $\Sigma$  and  $A$  is a constant multiplier matrix.
- 495.** In Bayesian model parameter estimation, if the observed data likelihood follows a Poisson distribution, identify the precise mathematical conjugate prior distribution for the rate parameter  $\lambda$ .
- 496.** A Markov clickstream process runs on three user states. The transition probability matrix is:

$$P = \begin{bmatrix} 0.4 & 0.4 & 0.2 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.3 & 0.6 \end{bmatrix}$$

Given an initial user probability state vector  $v_0 = [1.0, 0.0, 0.0]$ , compute the precise probability state vector after 2 step transitions ( $v_2$ ).

- 497.** Apply the union bound and Hoeffding's Inequality to express a general generalization error bound over a finite hypothesis space  $|H|$  containing  $M$  models.
- 498.** Write out the explicit joint probability distribution definition for a Dirichlet distribution with parameter vector  $\alpha = [a_1, a_2, a_3]^T$ .
- 499.** State the explicit optimization objective function for the Evidence Lower Bound (ELBO) used to approximate posterior fields in Variational Autoencoders (VAEs).
- 500.** Let  $Z_1, Z_2, \dots, Z_k$  be  $k$  independent standard normal variables. If you define a new metric summary  $W = \sum Z_i^2$ , name the resulting distribution family and state its analytical mean and variance.

### 3. STATISTICS (QUESTIONS 501–550)

#### A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 501–515)

- 501.** Calculate the sample mean and median for the following set of model cross-validation training times (in seconds):

**Data:** [10.5, 14.2, 11.1, 22.8, 13.9, 12.5, 15.4]

- 502.** Compute the sample variance and sample standard deviation for the training duration observations listed in Question 501. Use Bessel's Correction ( $n - 1$ ) explicitly.

**503.** Find the range and the Interquartile Range (IQR) for the following listing of deep network feature counts per layer:

**Data:** [32, 32, 48, 64, 64, 128, 128, 256, 2048]

**504.** Identify whether the final massive value (2048) in the feature counts array of Question 503 constitutes a mathematical outlier using the standard  $1.5 \times IQR$  rule.

**505.** A custom dataset sample of size  $n = 100$  yields a sample mean  $\bar{x} = 55.4$ . Given a known population standard deviation  $\sigma = 6.0$ , establish the 95% confidence interval for the true population mean.

**506.** Formulate the explicit Null ( $H_0$ ) and Alternative ( $H_1$ ) hypotheses required to test whether reducing data bit-resolution compromises model prediction accuracy metrics.

**507.** Define Type I and Type II errors precisely within the context of an automated spam-detection filter system.

**508.** An A/B test comparing user web engagement returns a calculated p-value of 0.006. Interpret this exact result relative to a strict significance threshold of  $\alpha = 0.01$ .

**509.** Compute the standard error of the mean (SEM) for a metric profile where the sample standard deviation is  $s = 3.15$  and the sample size is  $n = 81$ .

**510.** A categorical survey records the following customer sentiment responses across three classes. Compute the expected counts under the null hypothesis that responses are uniformly distributed:

**Satisfied: 70, Neutral: 35, Dissatisfied: 45**

**511.** Determine the directional skewness for a custom dataset where the calculated mean is 92.4, the median is 81.2, and the mode is 65.0.

**512.** Write out the mathematical steps required to assemble a sample covariance matrix for a two-dimensional feature array:

**Feature 1:** [2, 4, 6]  
**Feature 2:** [3, 5, 10]

**513.** Explain how the Central Limit Theorem allows you to perform hypothesis tests on the mean of an exponentially distributed population.

**514.** A 95% confidence interval for an NLP model's F1-score is calculated as [0.79, 0.85]. State the precise frequentist definition of this confidence interval.

**515.** What is the degrees of freedom for a sample variance estimator, and why do we divide by  $n - 1$  instead of  $n$ ?

## **B. INTERMEDIATE LEVEL (QUESTIONS 516–335)**

**516.** Perform a one-sample t-test: A baseline classification model takes 150ms to evaluate an image batch. A modified model version runs  $n = 16$  trials, yielding a mean latency of 144ms with  $s = 10ms$ . Calculate the t-statistic and state your conclusion for a two-tailed test at  $\alpha = 0.05$  ( $t_{crit} = 2.131$ ).

**517.** An independent two-sample t-test with equal variances compares two weight training initializations. Given the following data parameters, compute the pooled variance  $s_p^2$  and the resulting t-statistic:

**Init A:**  $n_1 = 10$ ,  $\bar{x}_1 = 5.40$ ,  $s_1 = 0.05$   
**Init B:**  $n_2 = 12$ ,  $\bar{x}_2 = 5.34$ ,  $s_2 = 0.06$

**518.** Construct a 99% confidence interval for the true difference between the initialization means using the data from Question 517 (assume  $t_{crit} \approx 2.845$  for  $df = 20$ ).

**519.** Calculate the Chi-Square ( $\chi^2$ ) test statistic for the following 2×2 contingency table tracking user conversions across two model recommendation strategies:

| Recommender | Converted | Ignored |
|-------------|-----------|---------|
| Model Alpha | 90        | 110     |
| Model Beta  | 120       | 80      |

**520.** Using the Chi-Square statistic from Question 519, determine if the choice of recommender significantly affects conversions at  $\alpha = 0.05$  (Critical  $\chi^2$  for  $df = 1$  is 3.841).

**521.** Compute the exact Pearson correlation coefficient  $r$  between feature  $X$  and target label  $Y$  from the following small training subset:

$X = [1, 3, 5, 7]$   
 $Y = [2, 6, 7, 9]$

**522.** Fit an ordinary least squares (OLS) linear trend line  $\hat{y} = \beta_0 + \beta_1 x$  using the small training sample from Question 521.

**523.** For the linear model derived in Question 522, compute the coefficient of determination  $R^2$  by calculating the ratio of the explained sum of squares (ESS) to the total sum of squares (TSS).

**524.** Explain the relationship between the statistical power of an experiment and the probability of making a Type II error. How does increasing sample size affect power?

**525.** A cluster response latency shows a log-normal profile with a mean log-latency of 4.2 and a log standard deviation of 0.3. Find the 95th percentile response latency threshold.

**526.** Convert the following coordinate pairs into a Spearman's rank correlation coefficient to assess non-linear monotonic correspondence:

$X = [12, 22, 19, 45]$   
 $Y = [150, 400, 310, 900]$

**527.** When analyzing user ratings that are highly skewed and ordinal, why is the Wilcoxon signed-rank test preferred over a standard paired t-test?

**528.** Complete the missing values in this one-way ANOVA framework. You are evaluating four model frameworks across 5 validations each. Given Between-Group Sum of Squares (SSB) = 80 and Within-Group Sum of Squares (SSW) = 120, compute the Mean Square fields (MSB, MSW) and the final F-statistic.

- 529.** Calculate Cohen's  $d$  for the experiment described in Question 516 to determine the effect size of the modified model version.
- 530.** A sample size of  $n = 25$  yields a sample mean of 18.5 and a sample standard deviation of 2.5. Construct the 95% confidence interval for the population mean using the t-distribution ( $t_{0.025, df=24} = 2.064$ ).
- 531.** Describe how the shape of the Student's t-distribution changes relative to the standard normal distribution as the degrees of freedom approach infinity.
- 532.** You run 50 parallel independent A/B tests on a website using a standard significance level of  $\alpha = 0.05$ . If the null hypothesis is true for all tests, what is the expected number of false discoveries? Apply the Bonferroni adjustment to find the corrected alpha threshold.
- 533.** Determine the numerator and denominator degrees of freedom for an F-test within an ANOVA design that compares 3 algorithms evaluated across 15 distinct datasets each.
- 534.** Define consistency as a property of an estimator. Is the sample variance  $s^2$  a consistent estimator for the population variance  $\sigma^2$ ?
- 535.** Write out the probability density function for the F-distribution and describe its structural role in testing variance equality between two populations.

### C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 536–550)

- 536.** Let  $x_1, x_2, \dots, x_n$  be i.i.d. random observations sampled from a Geometric distribution with success parameter  $p$ . Derive the Maximum Likelihood Estimator (MLE) for  $p$ .
- 537.** A user interacts with an ad portal, clicking the ad on the 3rd, 5th, 2nd, and 6th attempts across four independent sessions. Calculate the numerical MLE estimate for the click probability  $p$  based on your derivation in Question 536.
- 538.**  
Derive the Maximum Likelihood Estimator for the variance parameter  $\sigma^2$  of a normal distribution  $N(\mu, \sigma^2)$  given  $n$  i.i.d. observations, assuming the mean  $\mu$  is known.
- 539.** Derive the Maximum A Posteriori (MAP) estimator for the mean parameter  $\mu$  of a normal distribution given a normal prior  $\mu \sim N(\mu_0, \sigma_0^2)$  and  $n$  i.i.d. data points. Show how the MAP estimate balances prior knowledge with the empirical data.
- 540.** In a regression context, prove that the Mean Squared Error (MSE) can be decomposed into the sum of the square of the model bias, the model variance, and the irreducible noise variance.
- 541.** At a specific feature point  $x$ , a regression model has a true target value  $y = 12.0$ . Across several random training iterations, the model's predictions have an expected value of  $E[\hat{y}] = 11.0$  and a variance of  $\text{Var}(\hat{y}) = 0.5$ . If the irreducible error variance is  $\sigma^2 = 0.30$ , calculate the total expected MSE at  $x$ .
- 542.** Outline the step-by-step procedure for using the Bootstrap method to construct a 95% percentile confidence interval for the regularized weights of a Ridge regression model.

- 543.** A logistic regression model has parameter weights  $\mathbf{w} = [1.2, -0.8]^T$  and bias  $b = -0.5$ . For an input vector  $\mathbf{x} = [3.0, 2.0]^T$ , calculate the log-odds and the corresponding predicted probability for the positive class.
- 544.** Derive the Fisher Information matrix for a normal distribution's mean parameter  $\mu$  given  $n$  i.i.d. observations, assuming the variance  $\sigma^2$  is constant.
- 545.** Explain how changing the regularizer coefficient  $\lambda$  in Ridge regression scales the bias and variance of the model's parameters.
- 546.** Describe the primary assumptions of classical ordinary least squares (OLS) linear regression. What diagnostic plots or statistics can you use to check the independence assumption?
- 547.** Calculate the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) for a regularized neural network containing  $k = 15$  parameters evaluated over a sample of  $n = 400$  observations, given a maximized log-likelihood value of  $\ln L = -180.5$ .
- 548.** Why does evaluating a model using K-fold cross-validation provide a better estimate of generalization performance than evaluating it on the training set? Explain how cross-validation helps mitigate overfitting.
- 549.** If a regression model exhibits severe heteroscedasticity, how does this affect the validity of standard OLS hypothesis tests on the coefficients? Suggest a transformation technique to address this issue.
- 550.** Contrast the Benjamini-Hochberg procedure with the Bonferroni correction. Why is the Benjamini-Hochberg procedure generally preferred when performing high-dimensional feature selection across thousands of candidate variables?

## 4. DIFFERENTIAL CALCULUS (QUESTIONS 551–600)

### A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 551–565)

- 551.** Compute the first derivative of the polynomial loss curve function:  $f(x) = -3x^4 + 5x^3 - 8x + 12$ .
- 552.** Find the first derivative of the log-scaled function:  $f(x) = \ln(x^2 + 5)$  for all real numbers  $x$ .
- 553.** Apply the Product Rule to find the derivative of the combined scaling function:  $f(x) = x^4 \ln(x)$  for  $x > 0$ .
- 554.** Apply the Quotient Rule to find the derivative of the network ratio function:  $f(x) = e^x / (x^2 + 2)$ .
- 555.** Using the Chain Rule, find the explicit derivative of the composite radial activation function:  $f(x) = \exp(-2x^2)$ .
- 556.** Find the slope of the tangent line to the curve  $f(x) = x^3 - 3x^2 + 6$  at the explicit evaluation coordinate point  $x = 2$ .
- 557.** Evaluate the first derivative of the modified Sigmoid activation function  $g(x) = 1 / (1 + e^{-3x})$ . Express your final answer strictly in terms of  $g(x)$ .
- 558.** Find the partial derivatives  $\partial f / \partial x$  and  $\partial f / \partial y$  for the multi-variable function:  $f(x, y) = 3x^3y^2 - 5x^2y^4$ .

- 559.** Compute the gradient vector  $\nabla f(x, y)$  for the function  $f(x, y) = \ln(x^2 + y^2)$  at the operational evaluation point  $(2, 1)$ .
- 560.** Identify all critical points of the function  $f(x) = 2x^3 - 3x^2 - 12x + 5$  and classify each as a local minimum or local maximum.
- 561.** Compute the second derivative  $f''(x)$  of the function  $f(x) = x e^{-2x}$ . Find the point where the curve changes concavity ( $f''(x) = 0$ ).
- 562.** Find the derivative of the Exponential Linear Unit (ELU) activation function given by  $f(x) = x$  if  $x > 0$  and  $f(x) = 0.4(e^x - 1)$  if  $x \leq 0$ . Discuss its continuity at  $x = 0$ .
- 563.** Apply L'Hôpital's Rule to evaluate the numerical limit of the indeterminate ratio:  $\lim_{x \rightarrow 0} [(e^{2x} - 1) / x]$ .
- 564.** Compute the directional derivative of  $f(x, y) = 3x^2y^3$  at the point  $(1, 2)$  in the direction of the unit vector  $u = [1/\sqrt{2}, 1/\sqrt{2}]^T$ .
- 565.** Write out the first-order Taylor series approximation formula for a multi-variable function  $f(x, y)$  centered around a localized pivot point  $(a, b)$ .

## B. INTERMEDIATE LEVEL (QUESTIONS 566–585)

- 566.** Find the gradient vector  $\nabla f$  for the Mean Absolute Error (MAE) loss smooth approximation function for a single sample:  $f(w_1, w_2) = \sqrt{((w_1x_1 + w_2x_2) - y)^2 + \varepsilon}$  with respect to the weight parameter vector  $w = [w_1, w_2]$ .
- 567.** Evaluate the gradient of the function from Question 566 using the data values:  $x_1 = 2, x_2 = 1, y = 4, \varepsilon = 0.01$ , evaluated at current weights  $w_1 = 1, w_2 = 2$ .
- 568.** Perform one step of Gradient Descent to update the weight vector  $w$ . Use the current weights and gradient vector from Question 567 with a learning rate parameter  $\eta = 0.1$ .
- 569.** Compute the  $2 \times 2$  Hessian matrix  $H$  of second-order partial derivatives for the multi-variable optimization profile:  $f(x, y) = x^3 + 3xy + y^2$ .
- 570.** Analyze the eigenvalues of the Hessian matrix calculated in Question 569 evaluated at the critical point  $(3/2, -9/4)$  to determine its stability type.
- 571.** Find the derivative of the Swish activation function,  $f(x) = x \sigma(x)$ , where  $\sigma(x)$  is the standard Sigmoid function. Express your solution in terms of  $\sigma(x)$  and  $f(x)$ .
- 572.** Compute the derivative of the categorical cross-entropy loss function for a single sample class index:  $L = -y_i \ln(p_i)$  with respect to the model's predicted probability output  $p_i$ .
- 573.** Using the chain rule, compute the derivative of a regularized loss function  $L_{reg} = L_0 + \alpha|w|$  with respect to weight  $w$  for  $w \neq 0$ , given  $\partial L_0 / \partial w = G$ .
- 574.** Find the second-order Taylor series approximation for the exponential activation function  $f(x) = e^x$  centered around the evaluation pivot  $a = 0$ .

- 575.** Use implicit differentiation to find the derivative expression  $dy/dx$  along the hyperparameter boundary curve defined by  $x^2y + y^3 = 8$ .
- 576.** Compute the Jacobian matrix  $J$  for a coordinate mapping vector function:  $F(x, y) = [\ln(x) + 2y, e^{xy}]^T$ .
- 577.** Evaluate the Jacobian matrix  $J$  from Question 576 numerically at the sample coordinates  $(x = 1, y = 1)$ .
- 578.** Find the global minimum of the regularized cost function  $L(w) = (w - 4)^2 + \lambda|w|^2$  by optimizing analytically with respect to  $w$  when  $\lambda = 0.5$ .
- 579.** Given the vector function  $f(x) = \|x\|_2^2 = x^T x$ , write down the vector derivative expression for  $\nabla_x f(x)$ .
- 580.** State the definition of strict convexity for a multivariate function. If a function's Hessian matrix is positive semi-definite everywhere but not strictly positive definite, can a critical point still be a unique global minimum?
- 581.** Compute the derivative of the Parametric ReLU (PReLU) activation function:  $f(x) = \max(\alpha x, x)$  for both  $x > 0$  and  $x < 0$ , where  $\alpha$  is a learnable parameter.

### C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 582–600)

- 582.** Find the partial derivative of the GELU (Gaussian Error Linear Unit) approximation function  $f(x) = 0.5x(1 + \tanh(\sqrt{2/\pi}(x + 0.044715x^3)))$  with respect to  $x$  at  $x = 0$ .
- 583.** Determine the critical points of the function  $f(x, y) = x^4 + y^4 - 2x^2 - 2y^2$  and classify them using the Second Derivative Test.
- 584.** Explain how momentum modifies the standard gradient descent update formula. Write out the coupled system of equations for the velocity vector  $v_{t+1}$  and parameter vector  $w_{t+1}$ .
- 585.** Compute the partial derivative of the softmax function  $f(x_1, x_2) = e^{-x_1} / (e^{-x_1} + e^{-x_2})$  with respect to  $x_1$ .
- 586.** Using the method of Lagrange Multipliers, maximize the utility function  $f(x, y) = xy$  subject to the circular constraint  $x^2 + y^2 = 16$  in the first quadrant. Find the optimal values for  $x, y$ , and the multiplier  $\lambda$ .
- 587.** Consider a neural network computation path where an error term  $\delta^l$  at layer  $l$  is propagated backward. Derive the backpropagation matrix formulation for the error vector  $\delta^{l-1}$  at the previous layer given the weight matrix  $W^l$  and the activation derivative matrix  $\text{diag}(\sigma'(z^{l-1}))$ .
- 588.** For the multi-class Softmax activation vector where  $a_i = e^{z_i} / \sum e^{z_k}$ , and a cross-entropy loss function  $L = -\sum y_i \ln(a_i)$ , show that the total derivative of the loss with respect to the logit  $z_j$  simplifies beautifully to the error form:  $\partial L / \partial z_j = a_j - y_j$ .
- 589.** Derive the gradient with respect to matrix  $W$  for the linear layer transformation function  $f(W) = \text{Tr}(XWX^T)$  where  $X$  is a constant data matrix.
- 590.** Derive the gradient of the binary cross-entropy loss function summed over a mini-batch of  $N$  samples:  $L(w) = -(1/N) \sum [y_n \ln(\sigma(w^T x_n)) + (1 - y_n) \ln(1 - \sigma(w^T x_n))]$  with respect to the model weight vector  $w$ .

- 591.** Using the gradient vector from Question 590, derive the complete batch Hessian matrix expression with respect to  $\mathbf{w}$ . Show how this matrix can be written in the elegant form  $\mathbf{X}^T \mathbf{R} \mathbf{X}$  where  $\mathbf{R}$  is a diagonal weight matrix.
- 592.** Explain how the Levenberg-Marquardt optimization algorithm acts as a bridge between the Gauss-Newton method and standard Gradient Descent. Write down its regularized system update equation.
- 593.** Apply matrix differential rules to find the gradient of the log-likelihood function for a multivariate normal distribution with respect to its precision matrix (the inverse covariance matrix  $\mathbf{\Omega} = \mathbf{\Sigma}^{-1}$ ).
- 594.** Formulate the primal and dual optimization problems for a Support Vector Machine (SVM) classifier separating data with a soft-margin parameter  $C$ . Explicitly state the complementary slackness conditions.
- 595.** Derive the gradient of the trace of an inverted matrix function,  $f(\mathbf{A}) = \text{Tr}(\mathbf{A}^{-1} \mathbf{B})$ , with respect to the matrix parameter  $\mathbf{A}$ .
- 596.** In an LSTM cell, the cell state updates via  $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t$ . Derive the partial derivative matrix  $\partial \mathbf{c}_t / \partial \mathbf{c}_{t-1}$ . Use this to explain how the forget gate  $\mathbf{f}_t$  prevents vanishing gradients over long temporal sequences.
- 597.** Find the total differential  $dL$  for a system loss profile with structural dependency:  $L(\mathbf{u}, \mathbf{v}) = \ln(\mathbf{u}^2 + \mathbf{v})$  where  $\mathbf{u} = x e^y$  and  $\mathbf{v} = x^2 - y$ .
- 598.** Given the scalar cost function  $J(\mathbf{W}) = 0.5 \|\mathbf{W}\mathbf{X} - \mathbf{Y}\|_F^2$  where  $\mathbf{W}, \mathbf{X}, \mathbf{Y}$  are matrices, find the matrix gradient  $\nabla_{\mathbf{W}} J(\mathbf{W})$ .
- 599.** Prove that for a twice-differentiable multivariate function, a critical point is a strict local minimum if the gradient vector is zero and its Hessian matrix is strictly positive definite.
- 600.** For an optimization setup using AdaGrad, the parameter updates via  $\mathbf{w}_{t+1} = \mathbf{w}_t - (\eta / \sqrt{\mathbf{G}_t + \epsilon}) \odot \mathbf{g}_t$ , where  $\mathbf{G}_t = \sum \mathbf{g}_\tau^2$ . Show how the historical accumulation of partial derivatives modifies the effective local learning rate over long training runs.